

Unit 7.2 Acid Base Multiple Choice Practice

1. Nitrous acid, HNO_2 , has a pK_a value of 3.3. If a solution of nitrous acid is found to have a pH of 4.2, what can be said about the concentration of the conjugate acid/base pair found in solution?

- a. $[\text{HNO}_2] > [\text{NO}_2^-]$
- b. $[\text{NO}_2^-] > [\text{HNO}_2]$
- c. $[\text{H}_2\text{NO}_2^+] > [\text{HNO}_2]$
- d. $[\text{HNO}_2] > [\text{H}_2\text{NO}_2^+]$

2. A buffer solution can be formed by dissolving equal moles of:

- a. HF and NaOH
- b. KBr and Na_3PO_4
- c. CH_3COOH and NaCl
- d. HF and NaF

3. Which of the following changes would affect the pH of a buffer solution?

- I. Doubling the amount of acid and conjugate base used.
- II. Doubling the amount of water in the solution.
- III. Adding a small amount of strong acid or strong base.

- a. III only
- b. I and II only
- c. II and III only
- d. I, II, and III

4. Which of the following 1.0 M solutions would have a pH greater than 7.00?

- a. HCN
- b. KNO_3
- c. NH_4Cl
- d. $\text{NaC}_2\text{H}_3\text{O}_2$

5. A buffer solution may contain equal moles of

- a. a weak acid with its conjugate base
- b. a strong acid with its conjugate base
- c. a weak acid and a strong base
- d. a strong acid and a strong base

6. What is the conjugate acid and the conjugate base of HPO_4^{2-} ?

Conjugate Acid

Conjugate Base

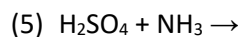
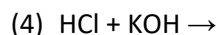
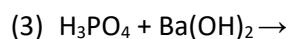
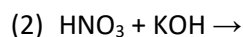
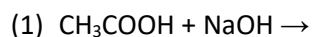
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|------------------------------|---------------------------|
| a. PO_4^{3-} | H_2PO_4^- |
| b. H_2PO_4^- | PO_4^{3-} |
| c. H_2PO_4^- | H_3PO_4 |
| d. H_3PO_4 | PO_4^{3-} |

7. When comparing 1.0 M solutions of bases, the base with the lowest $[\text{OH}^-]$ is the
- weakest base and it has the largest K_b value.
 - strongest base and it has the largest K_b value.
 - weakest base and it has the smallest K_b value.
 - strongest base and it has the smallest K_b value.

8. Which of the following pairs of substances would make a good buffer solution?

- | | |
|--|---|
| a. H_2SO_4 and LiOH | c. $\text{HC}_2\text{H}_3\text{O}_2$ and $\text{NaC}_2\text{H}_3\text{O}_2$ |
| b. HCl and KCl | d. HF and NaH_3 |

9. Consider the neutralization reactions between the following acid-base pairs in dilute aqueous solutions:



For which of the reactions above is the **net ionic** equation $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$?

- | | | | |
|---------|------------|---------|---------|
| a. 1, 3 | b. 2, 4, 5 | c. 2, 4 | d. 3, 5 |
|---------|------------|---------|---------|

10. What volume of 0.300 M KOH would be needed to completely neutralize 60.0 mL of 0.200 M H_2SO_4 ?

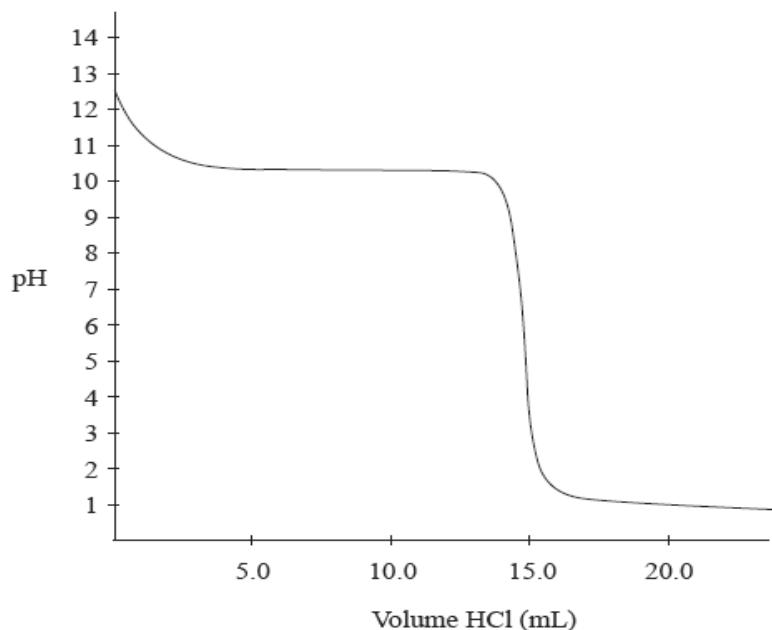
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| a. 40.0 mL | b. 60.0 mL | c. 80.0 mL | d. 100. mL |
|------------|------------|------------|------------|

11. Which of the following indicators would be most suitable for the titration of 0.10 M lactic acid ($\text{pK}_a = 3.08$) with 0.10 M $\text{KOH}(\text{aq})$?

- | | |
|---|--|
| a. phenol red ($\text{pK}_a = 6.9$) | c. thymol blue ($\text{pK}_a = 1.7$) |
| b. alizarin red ($\text{pK}_a = 4.5$) | d. methyl orange ($\text{pK}_a = 3.4$) |

Use the following information to answer questions 12–15.

A student titrates some 1.0 M HCl into 20.0 mL of methylamine (CH_3NH_2), a weak base which only accepts a single proton. The following titration curve results:



12. What is the concentration of the methylamine?
 - a. 0.50 M
 - b. 0.75 M
 - c. 1.0 M
 - d. 1.25 M

13. What is the approximate pK_b for methylamine?
 - a. 3.5
 - b. 5.5
 - c. 10.5
 - d. 12.5

14. What buffer region of this titration is located:
 - a. Below 3.0 mL
 - b. Between 3.0 mL and 14.0 mL
 - c. Between 14.0 mL and 16.0 mL
 - d. Above 16.0 mL

15. The methylamine is replaced by 20.0 mL of sodium hydroxide of an identical concentration. If the sodium hydroxide is titrated with the 1.0 M HCl, which of the following options accurately describes the pH levels at various points during the titration when compared to the pH levels at the same point in the HCl/methylamine titration?

Initial pH	Equivalence pH	Ending pH
a. lower	same	higher
b. higher	higher	same
c. same	higher	same
d. higher	lower	lower

16. What volume of 0.25 M HClO_4 would be needed to completely neutralize 30.0 mL of 0.20 M $\text{Ca}(\text{OH})_2$?

- a. 12 mL b. 24 mL c. 36 mL d. 48 mL

17. What volume of 0.100 M NaOH is required to completely neutralize 15.00 mL of 0.100 M H_3PO_4 ?

- a. 5.00 mL b. 15.00 mL c. 30.00 mL d. 45.00 mL

18. Which of the following titrations will always have an equivalence point at a $\text{pH} > 7.00$?

- a. weak acid with a weak base c. weak acid with a strong base
b. strong acid with a weak base d. strong acid with a strong base